

Output Form (OF)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: LAILA | Context: Arab High-School Arabic Essay Feedback (Qatar Primary)

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The output-form mismatch is the most direct and explicitly documented gap. LAILA evaluates models using QWK between human and model numeric scores [Q58] with hyperparameter optimization on average QWK [Q59, Q60, Q61]; LLMs are constrained to JSON-formatted scores via the outlines library [Q190]. The deployment requires open-ended natural-language revision suggestions in MSA accessible to G10–12 students [elicitation A3, regional YAML output_function]. QWK on numeric scores cannot capture pedagogical usefulness, actionability, or comprehensibility of generated feedback. The authors themselves recognize this limit by prohibiting high-stakes use [Q104] and recommending 'fairness auditing and stakeholder consultation before real-world application' [Q106]. Web research confirms no Arabic rubric-to-natural-language feedback evaluation framework exists [WEB-12, WEB-13, WEB-23]. OF was marked HIGH priority. Even basic infrastructure for evaluating feedback quality (rubrics for actionability, register-appropriateness, MSA fluency) is absent from the benchmark.

ID	Evidence
Q58	"We evaluate model performance using QWK to measure agreement between human and model scores." (p.8)
Q59	"Hyperparameters are tuned via Bayesian optimization with the Tree-structured Parzen Estimator algorithm (Bergstra et al., 2011), implemented using Optuna's TPESampler." (p.8)
Q60	"We ran 20 trials with 5 startup trials and a fixed random seed of 11." (p.8)
Q61	"The best configuration, selected based on average QWK on the development set, was used for final evaluation on the test data." (p.8)
Q104	"We explicitly prohibit its use for high-stakes educational decisions affecting individual students." (p.10)
Q106	"We recommend that any system developed using LAILA undergo fairness auditing and stakeholder consultation before real-world application." (p.10)
Q190	"For all the LLM experiments, we used vLLM for inference with the outlines library to enforce the JSON output format." (p.18)
WEB-12	AES literature explicitly identifies feedback generation as an overlooked area (https://link.springer.com/article/10.1007/s10462-024-11017-5)
WEB-13	AAEE Saudi system provides corrected essays but not trait-grounded revision suggestions (https://www.researchgate.net/publication/343796441_Automated_students_arabic_essay_scoring_using_trained_neural_network_by_e-jaya_optimization_to_support_personalized_system_of_instruction)
WEB-23	No Arabic system bridges trait scores to actionable natural-language feedback for secondary students (https://www.sciencedirect.com/science/article/abs/pii/S0306457319304935)

Strengths

- QWK is the established gold-standard metric for ordinal scoring agreement [Q58]
- JSON output enforcement and three-seed averaging for LLM experiments improve numeric-output reproducibility [Q189, Q190]
- Authors transparently flag the limits of the current evaluation [Q104, Q106]

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Q104	"We explicitly prohibit its use for high-stakes educational decisions affecting individual students." (p.10)

Q106	"We recommend that any system developed using LAILA undergo fairness auditing and stakeholder consultation before real-world application." (p.10)
Q189	"To account for variability in example selection, each experiment was repeated three times with random seeds 1, 11, and 42, and we report the average performance across runs." (p.18)
Q190	"For all the LLM experiments, we used vLLM for inference with the outlines library to enforce the JSON output format." (p.18)

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: No — benchmark output modality is structured numeric/JSON scores [Q58, Q181, Q190]; deployment requires open-ended natural-language MSA feedback. Direct mismatch.

OF-2: Not applicable — deployment is text-based; TTS not required.

OF-3: Literacy is high in target population (G10–12 students); MSA register-appropriateness for adolescents is the key accessibility consideration, on which the benchmark provides no signal.

OF-4: Documented form mismatch: QWK [Q58] cannot capture pedagogical usefulness, comprehensibility, or actionability of generated revision suggestions; benchmark explicitly prohibits operational deployment without independent validation [Q104, Q105].

ID	Evidence
Q58	"We evaluate model performance using QWK to measure agreement between human and model scores." (p.8)
Q104	"We explicitly prohibit its use for high-stakes educational decisions affecting individual students." (p.10)
Q105	"Users must commit to: (1) preventing re-identification attempts, (2) avoiding demographic profiling or inference, and (3) refraining from deployment in operational assessment contexts without independent validation." (p.10)
Q181	"In the zero-shot setting, the LLM is tasked with generating trait scores in JSON format, given the prompt text, essay, and trait rubrics." (p.18)
Q190	"For all the LLM experiments, we used vLLM for inference with the outlines library to enforce the JSON output format." (p.18)
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Information Gaps

- Whether any supplementary Arabic feedback-generation evaluation rubric is in development
- Whether downstream LLM-generated feedback can be calibrated against LAILA scores

Requires Expert Verification

- Pedagogical experts and Arabic-language teachers on what 'good' formative feedback looks like in MSA for G10–12

Remediation

Gap 1: Benchmark evaluates only numeric/JSON score agreement (QWK), providing no signal on natural-language feedback quality, actionability, or MSA register-appropriateness.

Recommendation: Construct a supplementary feedback-generation evaluation: pair each LAILA essay with teacher-authored model revision feedback, then evaluate generated feedback on a multi-axis rubric (faithfulness to trait scores, actionability, MSA register, student comprehensibility). Pilot with Arab high-school teachers and G10–12 students before any deployment.